

ORADEA, Romania

WMO: 150800

Lat: 47.0358N

Lon: 21.8958E

Elev: 137

StdP: 99.69

Time Zone: 2.00 (EUE)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-11.9	-9.0	-14.0	1.1	-10.9	-11.1	1.5	-7.6	10.6	3.5	9.7	3.0	2.3	110	0.495

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	12.2	33.8	21.2	32.0	21.0	30.1	20.4	23.3	30.0	22.2	28.9	21.3	27.9	2.9	190

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
21.1	16.0	26.4	20.1	15.0	25.0	19.1	14.1	23.9	69.7	29.6	65.6	28.9	62.5	28.1	27.4

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 9.0	7.9	6.9	-14.9	35.8	3.3	1.8	-17.3	37.1	-19.2	38.1	-21.1	39.2	-23.5	40.5		
(5)			-15.1	24.8	3.3	1.2	-17.5	25.6	-19.5	26.3	-21.4	27.0	-23.8	27.9		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	11.5	0.0	2.0	6.2	12.0	16.6	20.1	22.1	21.9	17.2	11.9	6.4	1.2		
	DBStd	8.89	4.88	5.21	4.67	3.93	3.58	3.46	3.15	3.37	3.76	4.38	4.72	4.85		
	HDD10.0	1126	309	226	132	24	2	0	0	0	1	32	127	274		
	HDD18.3	2858	567	457	376	191	76	21	6	7	65	204	358	530		
	CDD10.0	1681	1	2	14	85	207	302	374	370	216	89	19	2		
	CDD18.3	371	0	0	0	2	22	73	121	119	31	3	0	0		
	CDH23.3	3942	0	0	0	29	261	753	1287	1266	318	28	0	0		
	CDH26.7	1494	0	0	0	4	52	253	540	551	92	1	0	0		
(14) Wind	WSAvg	3.0	3.1	3.3	3.6	3.2	3.0	2.8	2.7	2.5	2.7	2.7	3.0	3.0		
(15) Precipitation	PrecAvg	600	33	32	34	46	55	74	73	50	55	50	48	50		
	PrecMax	880	66	99	141	85	135	160	167	135	152	110	90	120		
	PrecMin	359	12	1	5	8	14	13	8	1	10	2	0	0		
	PrecStd	138	16	22	32	19	29	37	36	35	36	31	22	29		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	13.0	15.9	20.8	26.2	30.0	33.6	36.1	36.2	32.1	26.0	20.2	15.1		
		MCWB	9.5	10.1	12.6	16.4	19.5	22.0	21.2	20.8	18.9	17.3	14.0	10.6		
	2%	DB	10.3	12.9	18.1	23.4	28.1	31.8	33.8	34.0	29.1	23.9	17.7	12.0		
		MCWB	7.9	9.0	11.4	14.9	18.5	21.5	21.3	21.0	18.2	16.2	12.7	9.2		
	5%	DB	8.8	10.8	15.8	21.5	26.2	29.9	31.9	32.0	27.0	21.8	15.1	9.8		
		MCWB	7.3	7.9	10.4	13.7	17.8	20.5	21.2	20.9	17.7	15.4	11.6	7.9		
	10%	DB	6.8	8.9	13.7	19.6	24.2	27.9	29.9	29.9	24.8	19.2	13.1	7.8		
		MCWB	5.8	6.9	9.2	12.9	17.0	20.0	20.7	20.4	17.3	14.5	10.8	6.5		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	10.0	10.8	13.8	17.3	21.2	24.0	24.9	24.7	20.9	18.6	15.0	11.5		
		MCDB	12.2	14.3	19.0	24.7	27.8	30.7	31.8	31.6	28.1	24.0	19.0	13.9		
	2%	WB	8.5	9.4	12.2	15.9	19.6	22.6	23.3	23.0	19.5	17.1	13.3	9.4		
		MCDB	10.1	12.3	16.7	21.9	25.9	29.2	29.8	30.1	26.1	21.7	16.5	11.7		
	5%	WB	7.1	8.3	11.0	14.7	18.5	21.5	22.2	21.9	18.6	15.9	12.2	7.9		
		MCDB	8.2	10.4	14.8	19.8	24.7	27.7	28.7	28.8	24.5	20.2	15.0	9.5		
	10%	WB	5.5	7.0	9.8	13.5	17.5	20.5	21.2	21.0	17.7	14.9	10.9	6.4		
		MCDB	6.5	8.6	13.2	18.3	23.1	26.4	27.9	27.7	23.1	18.7	13.0	7.5		
(35) Mean Daily Temperature Range	5% DB	MDBR	5.8	7.3	9.6	11.1	11.4	11.5	12.2	12.4	11.0	10.3	7.8	5.4		
		MCDBR	8.0	10.0	12.9	14.3	14.6	14.3	15.4	15.3	14.4	13.4	10.5	7.7		
	5% WB	MCWBR	5.9	6.7	7.5	7.1	6.8	5.9	5.5	5.2	5.7	6.7	6.6	5.6		
		MCWBR	7.2	8.9	11.6	12.4	13.1	12.8	13.3	13.6	12.2	11.7	9.4	7.0		
(40) Clear-Sky Solar Irradiance	taub		0.331	0.358	0.396	0.450	0.433	0.433	0.450	0.449	0.418	0.384	0.357	0.332		
		taud	2.324	2.280	2.208	2.080	2.194	2.228	2.187	2.187	2.240	2.317	2.340	2.352		
	Ebn at Noon		737	794	815	803	835	835	813	796	784	754	699	690		
		Edh at Noon	76	97	121	151	140	136	140	135	117	93	75	67		
(44) All-Sky Solar Radiation	RadAvg		1.07	1.91	3.17	4.53	5.51	6.00	5.89	5.37	3.76	2.47	1.37	0.82		
	RadStd		0.14	0.29	0.44	0.44	0.53	0.42	0.50	0.49	0.42	0.28	0.15	0.12		

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		+0.90	N/A	N/A	N/A	N/A	N/A	N/A	-194	+158	+106
(47) Regional (57 neighbors)		+0.69	N/A	N/A	+1.02	N/A	N/A	-127	-194	+113	+57

Nomenclature: See separate page